

ning dry all of a sudden started staring us in the face. We couldn't just hide behind layers of sunscreen anymore. Something more needed to be done.

Albert Einstein first explained the principle of photoelectric effect in 1905 – having to do with converting light



into electric current – getting a Nobel prize for his troubles. But it wasn't until 1954 that the blueprint of an efficient, modern-day photovoltaic cell was conceived in Bell Laboratories. At the heart of a photovoltaic or solar cell is silicon – the same substance used in computer chips – which is tasked with breaking down photons into electrons. It's not the most efficient way to be honest and it ain't cheap. Still, every country in the world is making a beeline towards solar power plants based on photovoltaic solar panels. Germany has the largest installation in the world generating 32.3 gigawatts, while China produces about one-fourth of all photovoltaic products in the world. India's very own Charanka Solar Park in Gujarat is currently the second largest solar power plant in the world.

But the photovoltaic cell is far from perfect. Scientists and researchers are constantly trying to find ways to boost its efficiency. Just last month, a team from MIT suggested that if silicon-based photovoltaic cells are coated with an organic dye called pentacene, it can dramatically increase power output to over 30 percent. More electricity from the same amount of sunlight. That has to be good news since these cells play such a crucial role in clean, renewable fuel in our quest for a greener future.

Plastic

While organizations fighting for environment preservation might not agree, plastic is one of the most important inventions ever. Wikipedia defines it as “any of a wide range of synthetic or semi-synthetic organic solids that are mouldable. Plastics are typically organic polymers of high molecular mass, but they often contain other substances.” The origin of artificial plastics is not completely clear but polystyrene, polyvinyl chloride (PVC) and rubber